

PROJECT COMPONENT LIST	
Category	Component & Purpose
Microcontroller & Core Modules	ESP32 Camera Module ► Main microcontroller with built-in Wi-Fi/Bluetooth and camera; handles processing, IoT communication, and image capture
	Arduino ► Secondary microcontroller for motor control logic, sensor interfacing, and low-level hardware management
	L298N Motor Driver Shield ► Dual H-bridge motor driver; controls direction and speed of DC motors on the robot chassis
	4WD Smart Robot Chassis Kit ► Mechanical frame with 4 wheels and DC motors; provides mobility platform for the fire-fighting robot
Sensors	Flame Sensor Module ► Detects infrared radiation emitted by fire/flame; triggers alarm and navigation response when fire is detected
	IR Sensor ► Infrared proximity/line sensor; used for obstacle detection or line-following navigation
Actuators / Alerts	Buzzer ► Emits an audible alarm when fire or hazard is detected; provides real-time audio alert
Power System	Li-ion Battery 3000mAh ► Rechargeable lithium-ion cell; primary power source for the robot and onboard electronics
	18650 Battery Holder ► Securely houses the 18650 Li-ion cells and provides electrical connections to the circuit
	18650 Battery Charger (AC) ► AC-powered charger for safely recharging 18650 Li-ion battery cells
Prototyping & Connections	Jumper Wires (pack of 20) ► Male-to-male/female wires for making temporary connections between components on breadboard/PCB
	MB Programmer ► Microcontroller programmer/flasher; used to upload firmware to the ESP32 or Arduino
	USB to TTL Adapter ► Converts USB signals to TTL serial; enables PC-to-microcontroller serial communication and programming
	USB to TTL Cable ► Serial communication cable for connecting and programming microcontrollers directly from a PC

PROJECT BUDGET			
Component & Purpose	Qty	Unit Price (₹)	Total (₹)
ESP32 Camera Module	1	995	995
Arduino	1	960	960
L298N Motor Driver Shield	1	220	220
4WD Smart Robot Chassis Kit	1	900	900
Flame Sensor Module	3	55	165
IR Sensor	2	45	90
Buzzer	1	20	20
Li-ion Battery 3000mAh	3	100	300
18650 Battery Holder	1	59	59
18650 Battery Charger (AC)	1	156	156
Jumper Wires (pack of 20)	20	4	70
MB Programmer	1	385	385
USB to TTL Adapter	1	100	100
USB to TTL Cable	1	320	320
GRAND TOTAL			4,740